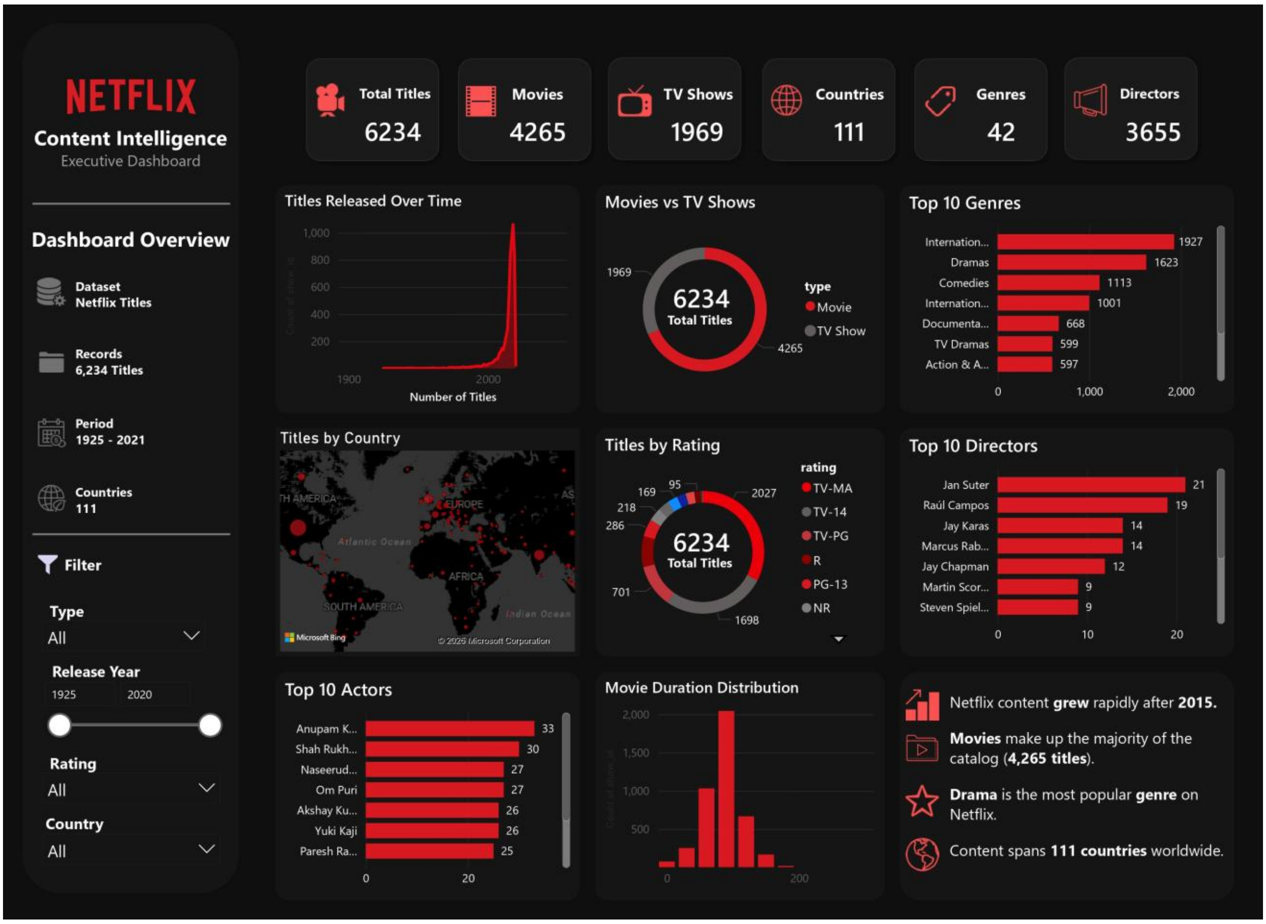
NETFLIX CONTENT INTELLIGENCE PLATFORM

Business Intelligence & Decision Support System

Final Report



Prepared By

Daksh Patel

Bachelor of Engineering (Information Technology)

Government Engineering College, Modasa

Executive Summary

The rapid growth of digital streaming platforms has generated enormous amounts of content-related data, making data-driven decision-making increasingly important for media companies. Netflix, one of the world's leading streaming platforms, offers thousands of Movies and TV Shows across numerous countries, genres, and audience categories. Managing such a vast content library requires effective Business Intelligence solutions capable of transforming raw metadata into meaningful strategic insights.

This project presents the development of a **Netflix Content Intelligence Platform**, an end-to-end Business Intelligence solution designed to analyze Netflix's publicly available content metadata. Rather than focusing solely on dashboard creation, the project emphasizes the complete data analytics lifecycle, beginning with raw data acquisition and progressing through data profiling, SQL-based preprocessing, Python-assisted normalization, database design, exploratory analysis, and interactive dashboard development.

The project utilizes Microsoft Excel for initial data understanding and quality assessment, MySQL for data storage and querying, Python for data normalization and preprocessing, and Microsoft Power BI for interactive visualization. Multi-valued attributes such as actors, directors, countries, and genres were transformed into normalized relational tables to improve analytical flexibility and database efficiency.

The final dashboard enables users to explore Netflix's catalog through interactive visualizations, providing insights into content distribution, release trends, ratings, genres, countries, directors, actors, and movie duration. Interactive filters allow users to perform dynamic analysis based on multiple business dimensions.

Overall, this project demonstrates the practical application of Business Intelligence, Data Analytics, SQL, Python, database normalization, and Power BI in solving real-world analytical problems and supporting strategic content analysis.

Table of Contents

1. Introduction

The entertainment industry has experienced a significant transformation with the rapid expansion of online streaming services. Millions of users consume digital content daily, resulting in continuously growing content libraries that require efficient organization and strategic management.

Netflix is one of the world's largest streaming platforms, offering thousands of movies and television shows across multiple genres, countries, and audience categories. As the volume of available content increases, understanding content trends becomes increasingly important for supporting strategic business decisions.

Business Intelligence provides organizations with the capability to convert raw data into meaningful information through data processing, visualization, and analytical reporting. Interactive dashboards enable decision-makers to monitor key performance indicators, identify trends, compare categories, and make informed decisions using historical data.

The objective of this project is to design and develop a comprehensive Business Intelligence platform that analyzes Netflix's publicly available metadata. The project demonstrates the complete analytics workflow beginning with raw dataset preparation and ending with an interactive executive dashboard.

Unlike many dashboard-only projects, this work includes data profiling, SQL processing, relational database normalization, Python automation, exploratory analysis, and Power BI visualization within a single integrated solution.

---

2. Business Problem

Streaming platforms continuously invest substantial resources in acquiring and producing new content. Without proper analysis, content investment decisions may rely heavily on assumptions rather than factual evidence.

Netflix maintains thousands of titles originating from different countries, directors, genres, and audience categories. Understanding this extensive content portfolio requires structured analysis capable of identifying trends, popular genres, regional distribution, audience preferences, and catalog growth over time.

Raw datasets alone cannot provide actionable business insights because they frequently contain missing values, inconsistent formatting, duplicated information, and multi-valued attributes that limit analytical flexibility.

Therefore, an integrated Business Intelligence solution is required to transform raw Netflix metadata into meaningful visual insights that support strategic decision-making.

---

3. Project Objectives

The primary objective of this project is to design and develop a complete Business Intelligence platform capable of analyzing Netflix's publicly available content metadata.

The specific objectives include:

• Perform detailed data profiling before analysis.

• Identify data quality issues including missing values and inconsistent formatting.

• Clean and preprocess the dataset using SQL.

• Normalize multi-valued attributes into relational database tables.

• Develop reusable Python scripts for automation.

• Perform exploratory data analysis.

• Design an interactive executive dashboard using Microsoft Power BI.

• Generate business insights from multiple analytical perspectives.

• Demonstrate an end-to-end Business Intelligence workflow suitable for real-world analytical projects.

---

4. Technology Stack

Technology	Purpose
Microsoft Excel	Initial data inspection and quality assessment
MySQL	Database storage and SQL analysis
SQL	Data cleaning, preprocessing, and business queries
Python	Data normalization and automation
Pandas	Data manipulation
SQLAlchemy	Database connectivity
Matplotlib	Exploratory visualizations

Technology	Purpose
Microsoft Power BI	Interactive dashboard development
GitHub	Project documentation and portfolio

5. Dataset Overview

The project utilizes the publicly available Netflix Titles dataset containing metadata for Movies and TV Shows available on the Netflix platform.

Dataset Source

Computational Thinking Laboratory

Netflix Titles Dataset

https://comp-think.github.io/laboratory/site/chapter/06/

Dataset Summary

Attribute	Value
Total Titles	6,234
Total Columns	12
Movies	4,265
TV Shows	1,969
Countries	111
Genres	42
Directors	3,655

The dataset contains descriptive information including title names, content type, directors, actors, countries, release years, ratings, durations, genres, and descriptions.

6. Project Workflow

The overall workflow followed during this project is illustrated below.

```
Raw Dataset
 |
 ▼
Excel Data Profiling
 |
 ▼
SQL Data Cleaning
 |
 ▼
Python Data Normalization
 |
```

▼
MySQL Relational Database
|
▼
Power BI Dashboard
|
▼
Business Insights

7. Data Profiling

Before beginning any cleaning or analysis, the Netflix dataset was carefully examined to understand its overall structure, completeness, consistency, and analytical readiness. Data profiling is an essential phase in Business Intelligence because it identifies potential quality issues that may affect dashboard accuracy and business insights.

The profiling process focused on evaluating missing values, duplicate records, data types, multi-valued attributes, and the overall suitability of the dataset for further analysis.

The dataset contains 6,234 Netflix titles distributed across Movies and TV Shows with twelve descriptive attributes. Several columns such as Director, Cast, Country, Rating, and Date Added contained incomplete information, while some attributes like Cast, Country, and Listed_In contained multiple values within a single cell. These characteristics indicated that normalization and preprocessing would be required before analytical reporting.

A detailed data quality assessment was initially performed in Microsoft Excel before beginning SQL preprocessing.

Column	Missing Values?	Needs Cleaning?	Notes
show_id	No	No	Unique identifier for each title.
type	No	No	Contains only Movie and TV Show categories.
title	No	No	Unique title names available.
director	Yes	Yes	Contains missing values that require preprocessing.
cast	Yes	Yes	Missing values and multiple actors in one field.
country	Yes	Yes	Missing values and multiple countries in one field.
date_added	Yes	Yes	Missing values; convert to DATE format.
release_year	No	No	Valid numeric field suitable for trend analysis.
rating	Yes	Yes	Few missing values requiring validation.
duration	No	Yes	Mixed format (minutes and seasons); requires transformation.
listed_in	No	Yes	Multiple genres stored in a single field; requires normalization.
description	No	No	Text field retained for future text analysis (NLP).

The profiling process also confirmed that the dataset was suitable for Business Intelligence after appropriate preprocessing and normalization.

---

8. Data Cleaning

After profiling, the dataset was imported into MySQL for systematic preprocessing. The objective of data cleaning was to improve overall data quality and prepare the dataset for normalization and visualization.

Several cleaning operations were performed during this phase.

Missing Value Analysis

Important attributes including Director, Cast, Country, Rating, and Date Added were checked for missing values. SQL queries were written to identify incomplete records before transformation.

Text Standardization

Leading and trailing spaces were removed from all important text columns using SQL functions to ensure consistency across the dataset.

Date Conversion

The Date Added attribute originally existed as a text field. A new DATE column was created, allowing proper chronological analysis inside Power BI.

Duration Transformation

Movie durations and TV Show seasons were originally stored together in a single text column. Separate numerical attributes were created to distinguish movie runtime from TV Show seasons.

These transformations significantly improved analytical flexibility while preserving the original dataset.

---

9. Database Design and Normalization

After completing data cleaning, the dataset was further optimized using **database normalization**. Several attributes in the original Netflix dataset contained multiple values within a single column, making direct analysis difficult and introducing data redundancy.

To overcome these challenges, the dataset was transformed into a relational database structure using dimension tables and bridge tables. This approach improved data consistency, reduced redundancy, and enabled efficient many-to-many relationships within Power BI.

The central table of the database is **clean_netflix_titles**, which stores the primary information for every Netflix title. Four normalized dimension tables were created to store unique entities:

- Actors
- Directors
- Countries
- Genres

Each dimension table is connected to the main dataset through an intermediate bridge table. This structure allows a single Netflix title to be associated with multiple actors, directors, countries, and genres while maintaining a normalized database design.

The final relational database consists of the following tables:

Table	Purpose
clean_netflix_titles	Master table containing cleaned Netflix titles
actors	Stores unique actor names
title_actor	Maps titles to actors
directors	Stores unique director names
title_director	Maps titles to directors
countries	Stores unique country names
title_country	Maps titles to countries
genres	Stores unique genre names
title_genre	Maps titles to genres

The normalized structure significantly improved analytical flexibility within Power BI by allowing independent filtering, aggregation, and relationship management across multiple business dimensions.

Figure 9.1: Relational database model showing the normalized schema consisting of the master table, dimension tables, and bridge tables used for Power BI analysis.

Relationship Summary

The final database model contains:

- **1 Master Table**
  - clean_netflix_titles
- **4 Dimension Tables**
  - actors
  - directors
  - countries
  - genres
- **4 Bridge Tables**
  - title_actor
  - title_director
  - title_country
  - title_genre

This relational model supports many-to-many relationships while minimizing data redundancy and improving dashboard performance.

---

10. SQL Analysis

SQL was used to transform raw data into meaningful business information through a series of analytical queries.

The project explored multiple business questions including:

- Distribution of Movies and TV Shows.
- Content ratings.
- Release year trends.
- Most active directors.
- Catalog growth over time.
- Top actors.
- Genre analysis.
- Country analysis.
- Movie duration analysis.

Each query was designed to answer a practical business question rather than simply retrieving raw records.

These SQL analyses later became the foundation of the interactive dashboard.

---

11. Python Automation

Python was used to automate repetitive preprocessing tasks that would have been difficult to perform efficiently using SQL alone.

The primary objective of Python was to normalize multi-valued attributes into relational tables suitable for Business Intelligence.

The normalization workflow consisted of the following steps:

- Reading cleaned data from MySQL.
- Splitting comma-separated values.
- Removing duplicate entries.
- Creating unique identifiers.
- Building bridge tables.
- Uploading normalized tables back into MySQL.

The same normalization logic was reused for Actors, Directors, Countries, and Genres by changing only the source column and destination table names. This reusable workflow reduced duplicate code while maintaining consistency throughout the ETL process.

Python was also used to generate exploratory visualizations that supported preliminary data analysis before dashboard development.

---

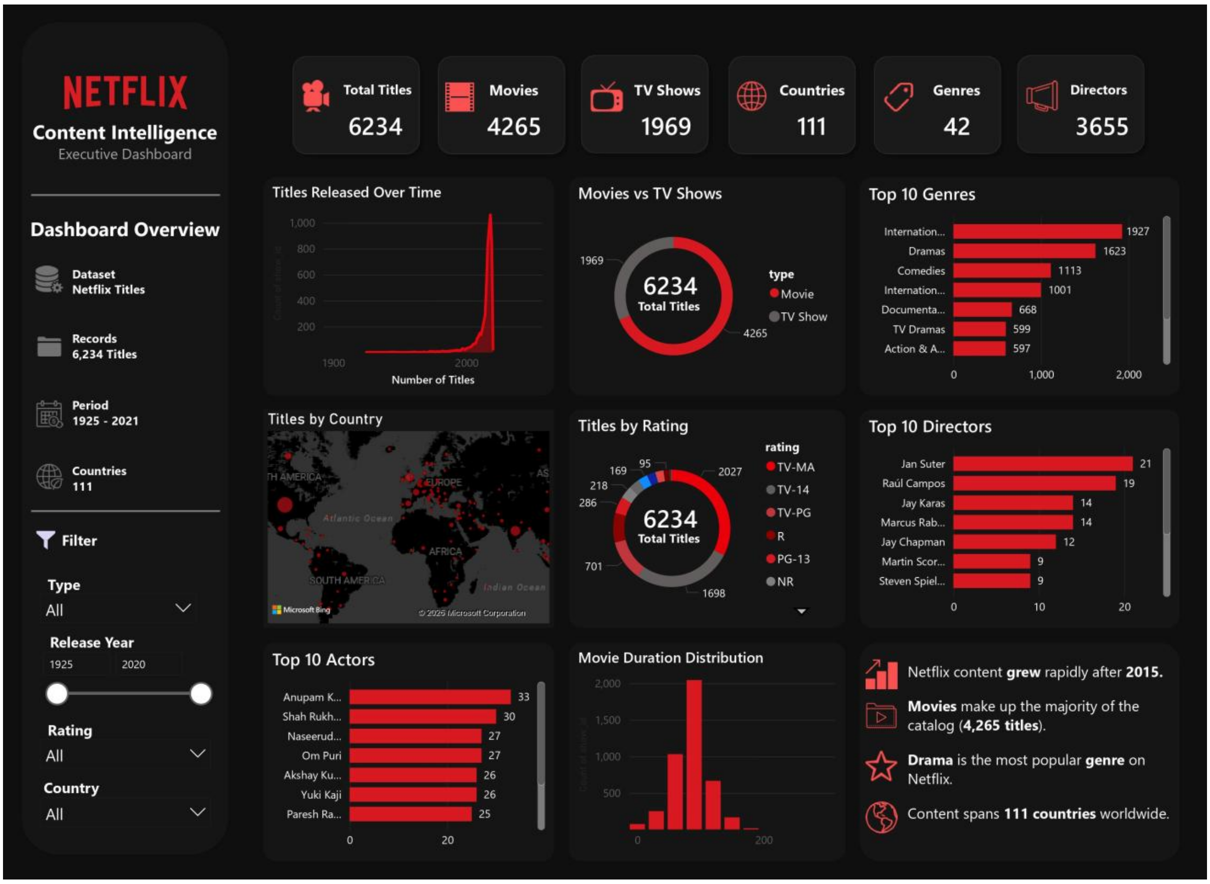
12. Power BI Dashboard Development

Following data cleaning, normalization, and SQL analysis, the processed dataset was imported into Microsoft Power BI to develop an interactive Business Intelligence dashboard. The objective of the dashboard was to transform analytical results into meaningful visual insights that could support business decision-making.

The dashboard was designed using a dark executive theme inspired by modern Business Intelligence applications. A consistent color palette consisting of black, dark gray, white, and Netflix red was selected to improve readability while maintaining the visual identity associated with Netflix.

Interactive filters were implemented to allow users to dynamically explore the dataset without modifying the underlying database. Users can filter the dashboard by content type, release year, rating, and country, enabling multiple analytical perspectives through a single interface.

The dashboard combines Key Performance Indicators (KPIs), charts, maps, trend analysis, and summary insights into one executive view. Every visual responds dynamically to user selections, allowing business users to explore Netflix's content portfolio interactively.



13. Dashboard Components

The dashboard consists of multiple analytical sections designed to answer different business questions.

13.1 Executive KPI Cards

The top section of the dashboard displays six key performance indicators that summarize the overall Netflix catalog.

The KPIs include:

KPI	Description
Total Titles	Total number of Netflix titles available for analysis
Movies	Total number of Movies
TV Shows	Total number of TV Shows
Countries	Number of unique producing countries
Genres	Number of unique genres

KPI	Description
Directors	Number of unique directors

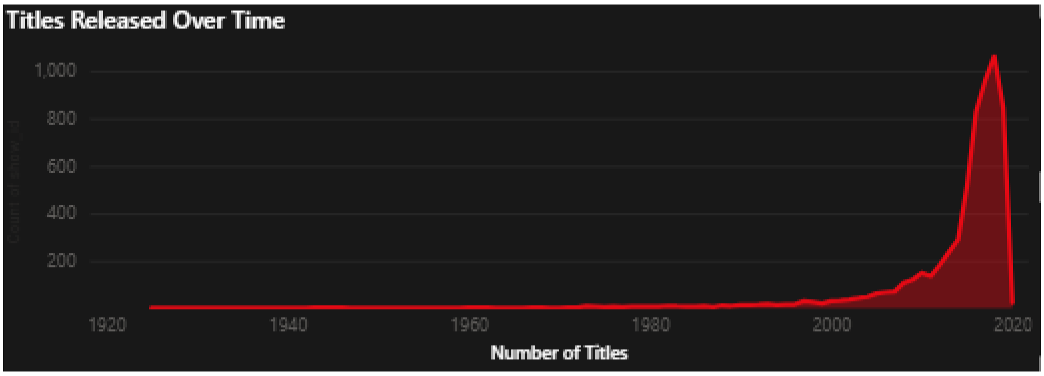
These KPIs provide decision-makers with an immediate overview of the content portfolio before performing detailed analysis.

13.2 Content Growth Analysis

The "Titles Released Over Time" line chart illustrates the historical growth of Netflix's content library based on release year.

The visualization enables users to identify periods of rapid content expansion and observe long-term trends in content production.

The dashboard indicates a significant increase in available titles after 2015, reflecting Netflix's rapid global content expansion strategy.

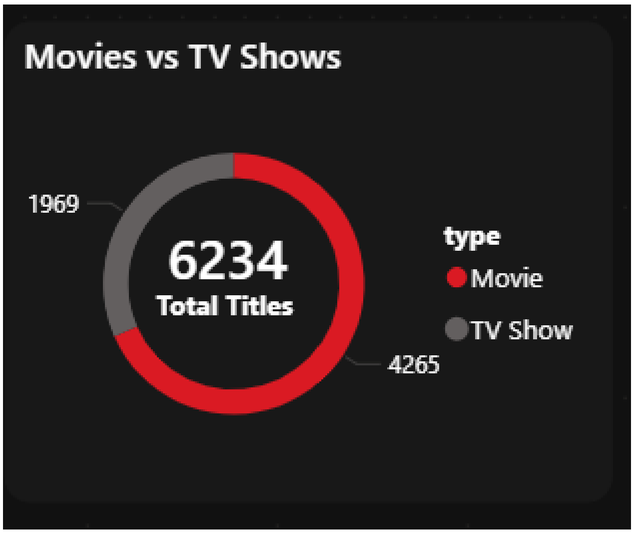


13.3 Movies vs TV Shows Distribution

A donut chart compares the overall proportion of Movies and TV Shows within the Netflix catalog.

The visualization quickly demonstrates that Movies represent the majority of Netflix's available content.

This chart allows business users to understand Netflix's overall content strategy at a glance.

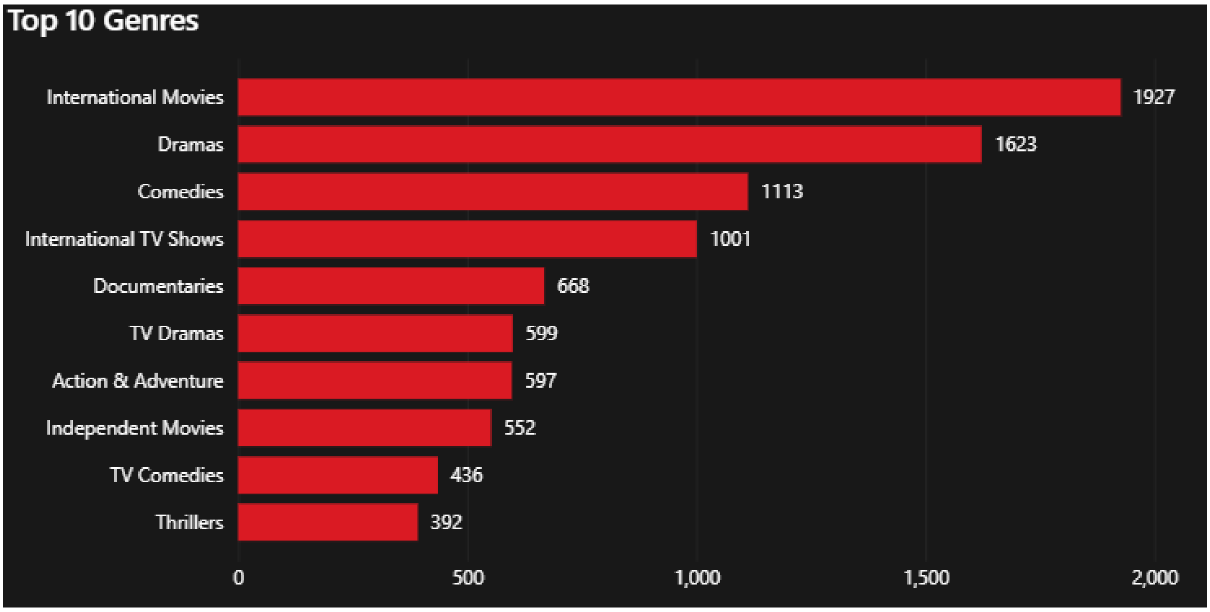


13.4 Genre Analysis

The Top 10 Genres visualization identifies the most frequently occurring genres available on Netflix.

This analysis helps identify content categories receiving the highest investment and audience availability.

The dashboard indicates that International Movies, Dramas, and Comedies dominate Netflix's catalog.



13.5 Country Analysis

The interactive world map illustrates Netflix's geographical content distribution.

Bubble size represents the number of titles produced by each country.

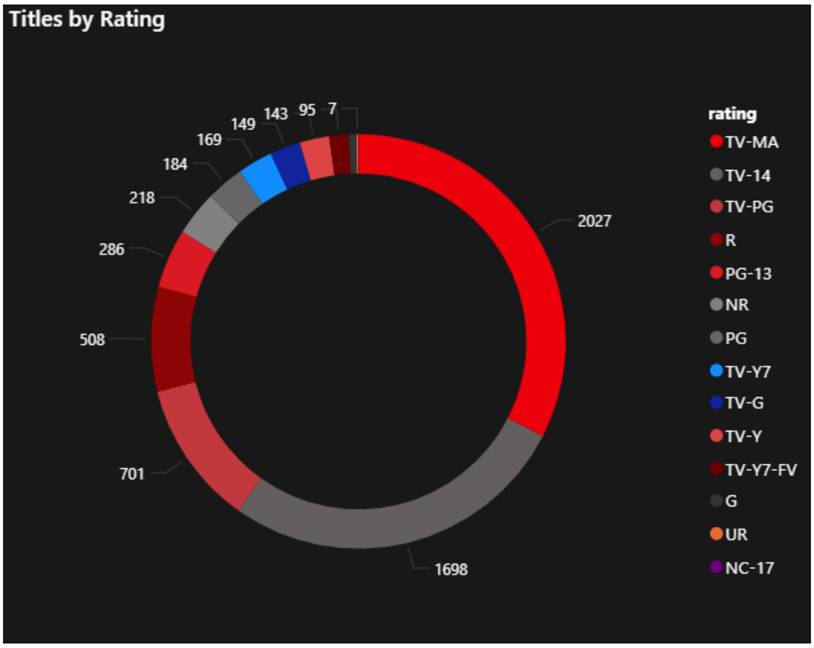
This visualization enables regional comparison and highlights countries with the largest contribution to Netflix's content library.

13.6 Rating Distribution

The Ratings visualization displays the distribution of audience certification categories across Netflix's content library.

This analysis helps identify the primary target audience segments served by Netflix.

The dashboard indicates that TV-MA and TV-14 represent the largest proportion of available titles.

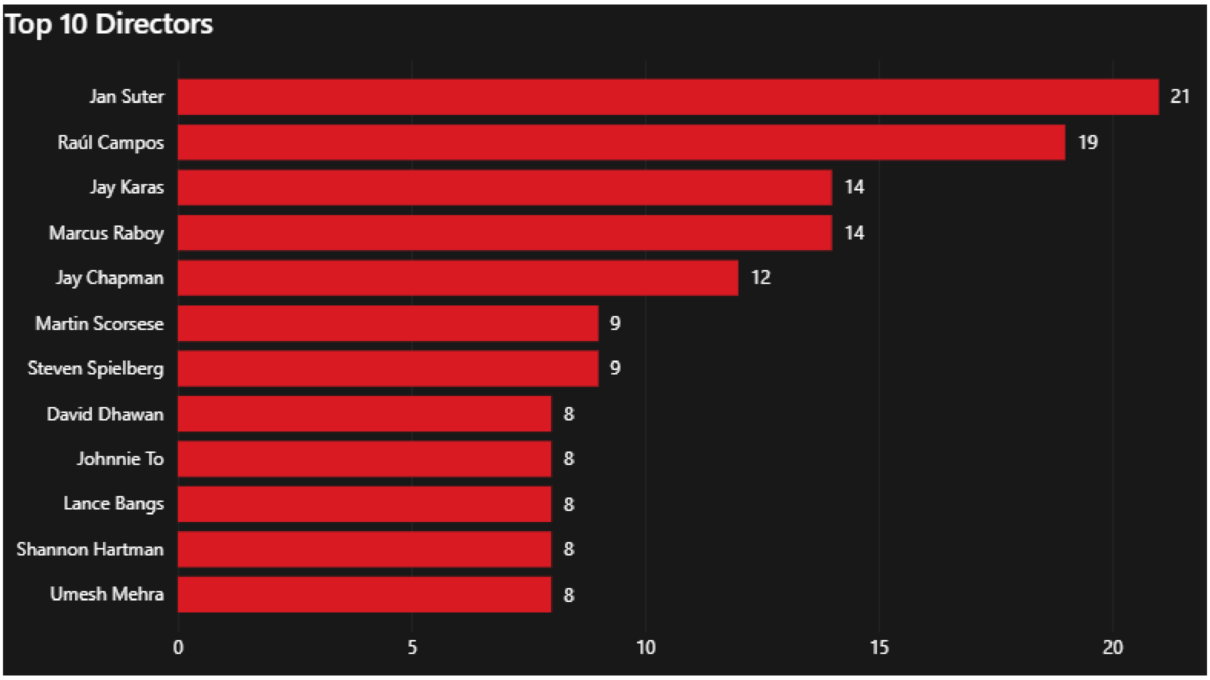


.

13.7 Director Analysis

The Top Directors visualization identifies directors with the highest number of titles available on Netflix.

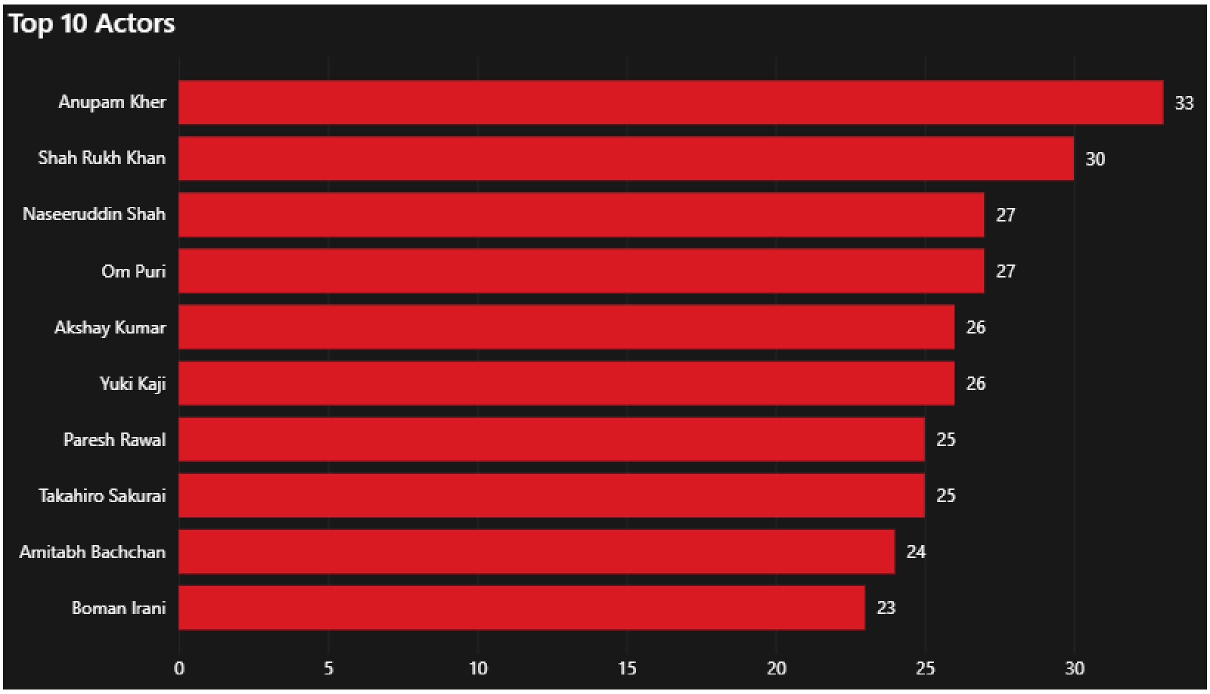
This analysis provides insight into recurring creative contributors and production partnerships.



13.8 Actor Analysis

The Top Actors visualization displays actors who appear most frequently within the Netflix catalog.

This information can assist in identifying highly represented performers across multiple productions.

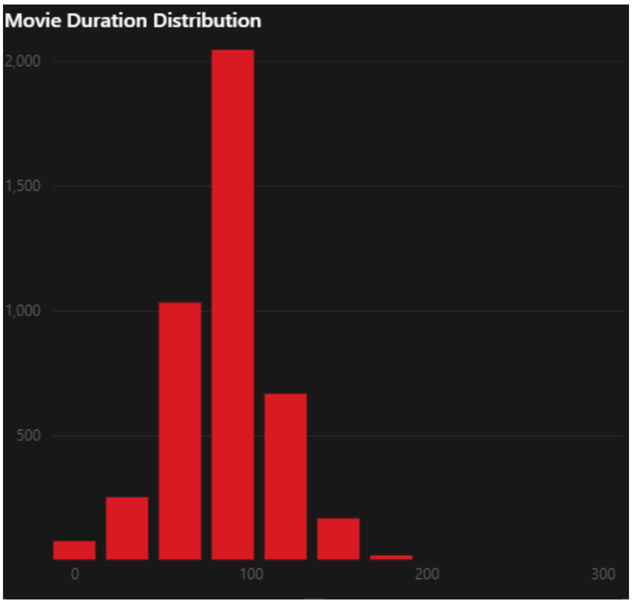


13.9 Duration Analysis

The Movie Duration Distribution histogram illustrates the frequency distribution of movie runtimes.

The visualization indicates that the majority of movies have durations close to approximately 90–120 minutes.

Understanding runtime distribution can support content categorization and production planning.



13.10 Interactive Filters

The dashboard contains four interactive slicers that allow users to customize the analysis dynamically.

Available filters include:

- Content Type
- Release Year
- Rating
- Country

These filters affect all dashboard visuals simultaneously, enabling interactive exploration without requiring additional SQL queries.

14. Business Insights

The completed dashboard generated several valuable business insights regarding Netflix's content portfolio.

Major observations include:

- Movies account for the majority of the Netflix catalog, representing more than two-thirds of all available titles.
- Netflix experienced rapid catalog growth after 2015, indicating a period of aggressive global expansion.
- International Movies and Dramas represent the largest content categories.
- TV-MA is the most common audience rating, suggesting a strong focus on mature audiences.
- Content originates from 111 different countries, demonstrating Netflix's global production strategy.
- Several directors have contributed multiple titles, highlighting long-term creative collaborations.
- The majority of movies have runtimes between approximately 90 and 120 minutes.
- Interactive filtering enables users to analyze trends across multiple dimensions without modifying the underlying database.

---

15. Key Findings

The analysis of the Netflix dataset revealed several significant patterns regarding the platform's content portfolio. By combining SQL analysis, Python preprocessing, and Power BI visualization, multiple business insights were identified.

The major findings of the project are summarized below:

Content Portfolio

- The dataset contains **6,234 Netflix titles**.
- Movies represent the majority of the catalog with **4,265 titles**.
- TV Shows account for **1,969 titles**, indicating Netflix has historically invested more heavily in movies.

Content Growth

- Netflix's catalog expanded significantly after **2015**.
- The release trend suggests an aggressive increase in content acquisition and production during recent years.

Genre Analysis

- **International Movies** is the largest genre within the dataset.
- Drama and Comedy are also among the most frequently occurring genres.
- Netflix maintains a diverse genre portfolio that serves multiple audience interests.

Audience Ratings

- **TV-MA** is the most common rating category.
- A large proportion of Netflix's catalog targets mature audiences.

Geographic Distribution

- Content originates from **111 countries**, demonstrating Netflix's extensive global presence.
- Multiple countries contribute to the platform's diverse international catalog.

Director Analysis

- Several directors have multiple titles available on Netflix.
- This indicates recurring collaborations with specific filmmakers.

Actor Analysis

- A number of actors appear repeatedly across the Netflix catalog.
- Such information can support talent analysis and future casting research.

Duration Analysis

- Most movies have a runtime between **90 and 120 minutes**.
- Runtime distribution follows expected commercial movie standards.

16. Business Recommendations

Based on the analytical findings obtained during this project, several recommendations can be proposed for strategic content planning.

Expand Successful Genres

International Movies, Dramas, and Comedies consistently dominate the content library. Continued investment in these genres is likely to maintain audience engagement while supporting global expansion.

Strengthen Regional Content Strategy

Since Netflix already receives content from 111 countries, further investment in regional productions may increase localization and improve subscriber growth across international markets.

Maintain Balanced Content Portfolio

Although Movies currently dominate the catalog, increasing the proportion of high-quality TV Shows could improve long-term viewer engagement due to episodic content encouraging continued platform usage.

Analyze Audience Preferences

The dominance of TV-MA suggests a strong mature audience base. Additional segmentation studies could identify opportunities for expanding family-oriented and children's programming.

Improve Metadata Quality

Missing values observed during profiling highlight the importance of maintaining complete metadata. Improving data quality would enhance future Business Intelligence reporting and recommendation systems.

Utilize Data-Driven Decision Making

The dashboard demonstrates how interactive analytics can assist decision-makers in monitoring content trends, evaluating production strategies, and identifying potential investment opportunities.

---

17. Challenges Faced

Several practical challenges were encountered throughout the development of this project.

Data Quality Issues

The original dataset contained missing values across multiple attributes including Director, Cast, Country, Rating, and Date Added. These issues required careful preprocessing before analysis.

Multi-Valued Attributes

Columns such as Cast, Country, Director, and Listed_In contained multiple values within single cells. These attributes required normalization to create an efficient relational database structure.

Database Design

Creating bridge tables and maintaining relationships between normalized tables required careful planning to ensure compatibility with Power BI.

Dashboard Design

Developing a dashboard that was both informative and visually appealing required balancing analytical detail with user-friendly design principles.

Interactive Filtering

Ensuring that all visuals responded correctly to slicers and maintained consistent filtering behavior required additional relationship management within Power BI.

---

18. Project Limitations

Although the project successfully demonstrates an end-to-end Business Intelligence workflow, several limitations remain.

- The dataset represents publicly available metadata rather than internal Netflix business data.
- Viewer engagement metrics such as watch time and subscriber retention were not available.
- Financial information, production costs, and revenue data were not included.
- Audience ratings represent content classifications rather than user satisfaction.

- Business recommendations are based on metadata analysis rather than operational business metrics.

These limitations provide opportunities for future enhancement using richer datasets.

---

19. Future Scope

Several improvements could further enhance the capabilities of the Netflix Content Intelligence Platform.

Machine Learning

Develop predictive models to estimate future genre popularity and content success.

Recommendation Systems

Integrate collaborative filtering or content-based recommendation algorithms.

Natural Language Processing

Analyze title descriptions using sentiment analysis and keyword extraction.

Real-Time Analytics

Connect Power BI directly to live databases or APIs for automatic dashboard updates.

Advanced Business Metrics

Incorporate subscriber growth, viewing duration, revenue, and production cost data to enable more comprehensive business analysis.

Cloud Deployment

Deploy the complete solution using cloud platforms such as Microsoft Azure or Google Cloud for scalable Business Intelligence reporting.

---

20. Conclusion

The **Netflix Content Intelligence Platform** successfully demonstrates the complete lifecycle of a Business Intelligence project, beginning with raw data and ending with an interactive executive dashboard.

The project integrates Microsoft Excel, SQL, Python, MySQL, and Power BI to transform publicly available Netflix metadata into meaningful business insights. Throughout the development process, data profiling, cleaning, normalization, exploratory analysis, and visualization techniques were applied to improve data quality and support informed decision-making.

The final dashboard provides users with an interactive environment for exploring Netflix's content portfolio through KPIs, charts, maps, filters, and summary insights. By combining multiple analytical technologies into a single workflow, the project demonstrates how Business Intelligence can convert raw information into actionable knowledge.

Beyond technical implementation, this project highlights the importance of structured data preparation, relational database design, and interactive reporting in solving real-world analytical problems. The methodologies presented in this project can be adapted to similar Business Intelligence applications across multiple industries.

Overall, the project successfully achieves its objective of developing a comprehensive Netflix Content Intelligence Platform capable of supporting strategic content analysis through data-driven decision-making.

---

21. References

1. Netflix Titles Dataset, Computational Thinking Laboratory
   https://comp-think.github.io/laboratory/site/chapter/06/
2. Microsoft Power BI Documentation
   https://learn.microsoft.com/power-bi/
3. MySQL Documentation
   https://dev.mysql.com/doc/
4. Python Documentation
   https://docs.python.org/3/
5. Pandas Documentation
   https://pandas.pydata.org/
6. Matplotlib Documentation
   https://matplotlib.org/
